



Module 21 Day 2: Introduction to Digital Forensics

Overview

Today's class will introduce the field of digital forensics. You will explore procedures for collecting, preserving, analyzing, and reporting forensic evidence.

Class Objectives

- Summarize the basic principles and methodologies of digital forensics.
- Describe various skill sets required for digital forensics jobs.
- Outline the proper approach to collecting, preserving, analyzing, and reporting forensic evidence.
- Conduct a preliminary review for a forensic case.
- Preserve and document evidence using Autopsy.

Lab Environment

Lab Details

User: kali

Password: kali

In this module, you will use a new Forensics lab environment within bootcampspot.

01. Welcome and Overview

Digital forensics is a very exciting cybersecurity field that requires a unique set of skills.

- Recall that penetration tests attempt to discover vulnerabilities to prevent breaches and cybercrimes.

- We know that not all defenses can withstand the constant network attacks of today's cyber landscape.
- Therefore, digital forensics attempts to discover evidence of a crime **after** it's been committed.

Roles, Knowledge, and Skill Sets

Digital forensics professionals must thoroughly understand hardware, operating systems, and computer networks.

- The following link is from NICCS detailing skills, knowledge, and other aspects associated with digital forensics:
 - [Cyber Security Workforce Framework: Digital Forensics](#)

Roles and responsibilities in digital forensics are not well-defined across job titles. Specific job roles and responsibilities vary by organization, but may include the following:

- **Computer forensics investigators:** These professionals gather digital information for computer system investigations, producing criminal evidence that can be used in a court of law.
 - Legal cases involve data theft, espionage, and cyberwarfare.
 - This field involves analyzing data found on hard drives, network communications, and cloud-based data communication systems, as well as producing reports and possibly testifying in a court of law.
- **Computer forensics technicians:** These are digital detectives who work with law enforcement or government entities or as private investigators.
 - They use investigative and computer analysis techniques to acquire, analyze, and preserve digitized evidence.
 - They also inspect storage media and retrieve corrupted and deleted files from computing storage media, such as hard drives, removable flash media, and mobile devices. This material is used in legal proceedings.
- **Forensic computer analysts:** These professionals specialize in recovering emails or other data that has been encrypted or deleted. This material is used in legal cases involving computer crimes.

- Evidence produced by forensic computer analysts is used by law enforcement to assist with ongoing investigations.
 - Evidence may also be used in a court of law to help convict cybercriminals.
-

02. Introduction to Digital Forensics

What Is Digital Forensics?

Digital forensics is a field dedicated to identifying, extracting, preserving, and reporting on information obtained from computer and network systems.

- The following definition of digital forensics was published in the article ["Digital Records Forensics: A New Science and Academic Program for Forensic Readiness"](#) in the *Journal of Digital Forensics, Security and Law*:
“The use of scientifically derived and proven methods toward the preservation, collection, validation, identification, analysis, interpretation, documentation and presentation of digital evidence derived from digital sources for the purpose of facilitating or furthering the reconstruction of events found to be criminal, or helping to anticipate unauthorized actions shown to be disruptive to planned operations.”

Legal Requirements

Why is it important to preserve the integrity of digital evidence?

Forensic evidence must satisfy a set of legal standards to be admissible in court.

- Imagine spending months searching through computer and network systems collecting evidence, only to have it thrown out due to improper procedures.
- Forensic evidence is held to the same standards as any other evidence submitted in a legal case. All evidence must be wholly intact and unaltered from the scene of the crime.
- Once evidence is collected, the investigator or analyst is accountable for its ownership and whereabouts at all times.

Chain of Custody

A chain of custody is a documentation of the possession of evidence.

- Evidence taken into custody must be accounted for. This means that its whereabouts and ownership must be accurately documented at all times, from the time it is acquired to when it is presented in the courtroom.
- When testifying about the integrity of evidence, one must be able to provide a documented, uninterrupted chain of custody that includes:
 - A detailed log of each person who accessed or handled the evidence
 - The purpose of that access
 - Dates and times it was handled
- If you are not able to show an uninterrupted chain of custody, months or even years of hard work can be considered inadmissible, leading to the possible release of criminals. This is why it's so critical to ensure that a proper chain of custody is maintained at all times.

The Digital Forensics Process

The National Institute of Standards and Technology defines a process for performing digital forensics in [Special Publication 800-86](#) as follows:

- **Collection:** We must first collect the data before we can examine and analyze it. The collection phase is the springboard of the digital forensics process, which includes identifying, labeling, recording, and acquiring data from sources while following procedures to preserve data integrity.
- **Examination:** After we've collected the evidence, we begin the examination phase, which ensures that all data collected is relevant to the case. This includes forensically processing collected data and assessing and extracting data of interest while preserving the integrity of the data. This usually means working from a copy rather than the original.
- **Analysis:** After we've identified all relevant evidence, we begin analyzing the results of the examination, using legally approved methods and techniques, to derive information that addresses the questions that inspired the collection and examination.
- **Reporting:** After we've analyzed all of the evidence, digital forensics investigators are required to formally report the results of the analysis, which may include describing the actions used, explaining how tools and procedures were selected, determining what other actions need to be performed, and recommending improvements to the forensic process.

Evidence Acquisition

Evidence acquisition is one of the most critical steps in the entire forensic process. If done improperly, it can result in evidence being missed, lost, or deemed inadmissible.

- Digital forensics data has two classifications: Network-based or host-based.
 - Network-based data comes from data communications captured by network-based systems such as an IDS, IPS, and firewalls, in the form of a packet capture or similar.
 - Packet captures are useful for reconstructing events involving computer break-ins.
 - Logs from firewalls also provide insight into network activity.
 - Host-based data is typically found on a system that encompasses various artifacts.
 - Investigations involving computer break-ins may involve the forensic examination of local file systems, programs, access to critical documents, and/or alterations to system files and directories.
 - Host-based examinations may involve reconstructing internet use, unauthorized activity, email recovery, and identifying malicious files.

Creating Forensic Backups

During an investigation, it is important that evidence is not contaminated. As a forensic investigator, you will image the data. This image, not the original media, will be used for analysis.

- A bit-stream image captures all created sections of a hard drive, known as partitions, whether used or not, and all unallocated drive space that doesn't belong to partitions.
 - This method allows forensic examiners to recover deleted files and fragments of data that may exist on a hard drive.
 - A local file system backup is inadequate for forensic analysis. If you make a copy through the file or operating system, you can see only the data that the operating system sees. This won't capture deleted files or slack space, which may contain relevant data. Therefore, you must obtain a bit-level copy.
-
- A bit-level copy or bit-stream image copies over everything: Slack space, deleted files, and regular files.
 - A local backup only copies the files that are on the system. This does not copy over slack space or deleted files.

- Slack space is the area of a file system that indicates deleted or truncated files.

Local backups of hard drives should be performed only if the machine can be taken offline. If it's taken off of the network, then the owner of the machine cannot send remote commands to, for example, wipe the computer or lock forensics experts out.

- A forensic backup differs from a typical local backup:
 - Typical local backups target intact files. The backup image does not include the free space and slack space on a hard drive.
 - Forensic backups (system image or bit-level backups) create an exact replica of all contents of the entire hard drive, including slack space, free space, and deleted files.
 - Forensic backup images are created in the following formats:
 - Raw formats: These formats can be created with programs such as dd, ddfldd, and ddcdd.
 - .bin
 - .dd
 - .img
 - .raw
 - Advanced Forensic Format (AFF): The AFF format is for disk images and related forensic metadata.
 - .AFF
 - .AFF4

Working with Live Systems

Always use caution when working with live systems. An attacker might be waiting for the user to log back into the system to complete the attack.

- The primary reason for working on a live system is to capture items that will not survive a power loss, such as volatile memory, swap files, running processes, and active network connections.
- Explain that mistakes such as writing data to memory or disk can potentially destroy evidence, which is why it is critical to use forensic tools such as write blockers.
 - A write blocker is a device that allows anything connected to it to perform only read operations, preventing the drive from being written to and overwriting evidence.

File Systems

You may encounter the following file systems during an investigation.

- New Technology File System (NTFS): Supported under Windows 10, 8, 7, Vista, XP, and NT
- File Allocation Table (FAT): Supported by older and newer versions of Windows
- Apple File System for 10.13+ is APFS
 - Apple File System for 10.12- is MacOS extended
- Fourth Extended File System (Ext4): Used in most Linux distributions

Overview of Storage Media

- Physical vs. virtual drives
 - A virtual drive resides on a physical drive and emulates the characteristics of a hard drive. The drive can be mounted and dismounted and data can be read and written to it.
 - In general, drives are limited by the speed at which they can receive and send information.
- Mechanical hard drives
 - Mechanical hard drives have very delicate moving parts that can be damaged if not handled properly.
 - They can have far larger storage capacities than other drives. Therefore, the imaging process can take an extremely long time when performing a bit-level copy.
 - Implications for forensics: Data can be recovered from badly damaged or compromised devices. This requires extensive knowledge of the inner workings of hard drives and how they store data.
- Flash storage
 - Flash storage devices have no moving parts and use flash memory, allowing for quicker read–write access.
 - Flash memory or flash storage is non-volatile, meaning it holds data even when it's not connected to power. This technology is used in devices such as USB drives, mobile phones, cameras, and tablet computers.
- Solid-state drives (SSDs)
 - SSD storage devices use flash memory chips and have no moving parts. They also have larger storage capacity and have faster read–write access.
 - Implications for forensics: An SSD is not a mechanical drive. One must be careful with the forensics tools used to image and recover data. The data on an SSD can be lost or wiped in seconds.
- SD or microSD cards
 - SD cards store data inside a flash memory chip, similar to SSDs. They are used in mobile phones and smartphones.

- Implications for forensics: It is possible to retrieve data from an SD card even if it has been reformatted or if the data has been deleted. This is because data is not immediately erased, the storage is only set aside for reuse. As long as the storage space has not been used, the preexisting data is still available. This is very useful for forensic recovery.

You may encounter password-protected storage media and password-protected firmware.

- Firmware is a specific class of computer software that provides low-level control for a device's specific hardware.
- These situations present extreme challenges to forensic investigators and must be considered during the initial stages of the investigation.

Working with Cloud-Based Evidence

Working with online evidence, such as hard drives and volatile memory located on cloud-based hardware, presents a new set of challenges.

03. Activity: Digital Forensics in the Cloud

- In this activity, you will explore a case concerning a DoS attack on an online shopping website.
- You will list at least four challenges associated with conducting a cloud forensics investigation.
- You will also research how these challenges affect the chain of custody during a digital forensics investigation.
- As a bonus, you will be asked to provide at least one example of how evidence can be obtained to validate the chain of custody in cloud-based forensic investigations.

- [Activity File: Digital Forensics in the Cloud](#)

04. Activity Review: Digital Forensics in the Cloud Activity

05. Overview of Digital Forensics Types and Methodology

In addition to cloud-based forensics, digital forensics is a continuously evolving scientific field that incorporates many subdisciplines.

Note some of the following subdisciplines:

- Computer forensics is the identification, preservation, collection, analysis, and reporting of evidence acquired from computers, notebooks, and storage media. It supports investigations and legal proceedings.
- Disk forensics involves acquiring and analyzing information stored on physical storage media, such as hard drives, smartphones, GPS systems, and removable media. This involves recovering hidden data, deleted data, and slack space information.
- Memory forensics inspects computer memory to identify an attacker's activities on a system.
 - This area requires the broadest skill set. Professionals must understand CPU architectures, operating systems and memory management, page tables, and virtual addressing, among other things.
 - Memory forensics is different from disk forensics in that the data investigated is stored in volatile memory, such as RAM. Volatile memory refers to storage media that loses its data when rebooted. Data on hard drives does **not** change between reboots, so it is non-volatile.
- Network forensics is the monitoring, recording, storing, and analysis of all network traffic to determine the source of security events.
 - Can you think of any tools that can be used for network forensics?
 - Network forensic tools include Wireshark, NetworkMiner, and Snort, among others.
 - Investigators in this area have an excellent understanding of communication and network protocols and the tools needed to capture and analyze data.

- Email forensics analyzes the source and content of an email as evidence. Email forensics includes the process of identifying the sender, recipient, date, time, and original location of an email message.
 - Can you think of any famous email forensics cases?
Hint: What about the 2016 presidential election? Sony?
 - Email forensics is important when providing attribution, as is often required in conspiracy and fraud cases.
- Mobile device forensics involves recovering digital evidence from smartphones, GPS devices, SIM cards, PDAs, tablets, and game consoles.
 - For example: Cell phone forensics can be used in a distracted driving case. A forensic expert can use this data to analyze what was happening at the time of the accident.
- Other forensic areas include drone, internet, and malware technologies.

Digital Forensic Investigation Methodology

Now that we understand the importance of maintaining an accurate chain of custody, we will explore the methodology for conducting digital forensic investigations.

A successful digital forensics investigation is crucial for the prosecution of computer criminals.

- It is crucial to have a clearly defined, step-by-step process for collecting digital evidence to ensure it is presentable in a court of law.

Digital forensics has several investigative frameworks. They include variations of the following phases:

1. Prepare for the investigation
2. Collect the evidence/forensic recovery
3. Preserve the evidence
4. Electronic discovery and analysis
5. Present and report

1. Prepare for the Investigation

Preparation is an important first step in conducting an investigation. However, there is not always a lot of time for it.

- As an investigator, you must consider whether the incident was remote or local, what laws are relevant, and what tools should be used (GUI or CLI).

2. Collect the Evidence/Forensic Recovery

The success of the analysis phase depends on the collection phase.

- During the collection phase, the investigator makes decisions about what data to collect and the best way to collect it.
- During forensic recovery, the evidence is extracted from a device and a master copy is made.
- How you collect the evidence determines if the evidence is admissible in court.

3. Preserve the Evidence

Investigators never work on the original copy of the evidence. A read-only master is made and stored in a digital vault.

- All processing is done on the copy.
- Investigators use a write blocker to ensure the prevention of data contamination.

A cryptographic digest or hash is made to ensure that evidence has not been altered in any way.

- This hash ensures the integrity of the evidence throughout the investigative process.

4. Electronic Discovery and Analysis

Analysis is done after data has been collected. This is sometimes referred to as dead analysis.

- It's important to document everything, including:
 - Time
 - Date
 - Locations
 - Applications used
- Additionally, your evidence must be reproducible. If it is not, it may be ruled as inadmissible.

5. Present and Report

In this phase, investigators write an expert report that lists all tests conducted, when and how they were conducted, what was found, and the investigation's conclusions.

- Forensics investigators may be called to testify as expert witnesses in trials or depositions.
-

06. The Pure Gold Credit Union Case

- Over the next two days, you will work on a case involving a fictitious bank theft at Pure Gold Credit Union.
 - Employees at Pure Gold Credit Union conspired to steal funds from their credit union's branch.
 - Suspicious activity was reported to senior management and devices were seized, including physical images of mobile devices.
 - The seized evidence was processed by the ingest team at the Crime Laboratory.
 - The evidence was backed up using the EnCase forensic application.
 - In the next activity, you will begin to review and document their investigation.
 - You will analyze an iPhone image of one of the employees. This will test your ability to use digital forensic tools to perform a scientific investigation.
-

07. Activity: The Pure Gold Credit Union Case

- In this activity, you will work as forensics investigators investigating the Pure Gold Credit Union Case.
 - You will work in small groups to become familiar with the case.
 - You are provided a list of the evidence for the case.
 - You are tasked with answering several questions while reviewing the case.
- [Activity File: Pure Gold Credit Union Case](#)
 - [Resource File: Pure Gold Credit Union Scenario PDF](#)
-

08. Activity Review: The Pure Gold Credit Union

09. Introduction to Autopsy

The last page of the Pure Gold Credit Union Activity describes the storage devices that were imaged as evidence.

- The seized evidence included an iPhone image. This image contains storage devices that hold operating system data, file system data, applications, documents, pictures, videos, and music.
- We'll be looking for evidence related to the case, including:
 - SMS messages that show intent to commit a crime.
 - IP address information that indicates GPS location data.
 - Emails and email addresses that contain incriminating evidence.

Open-source Tools

Digital forensics relies on examiners to analyze and interpret data retrieved by trusted forensics examination tools.

- You can perform a complete investigation using only open-source tools.

You will be using the [Sleuth Kit Autopsy Forensic Application](#).

- Autopsy is an open-source graphical tool that runs on Windows and Linux platforms.
- We'll be using Autopsy in the Kali Linux operating system.

To run Autopsy properly, we'll need to ensure that the Oracle Java Development Kit is installed and updated on our Kali machine.

- Run the following in a new terminal window: `sudo apt install -y default-jdk`.
 - The password is 'kali'

Prepare the Data

- In the real-world, we would first run a virus scan on the image. This takes time, so we won't be doing it during class. If any virus files are found during the scan, they should be documented and included in our report.
- Next, we'll generate MD5 and SHA-256 hashes for the evidence. This will ensure the integrity of the evidence. It's best practice to hash the evidence file using two different hashing algorithms. One can be used as a backup hash for the other.
 - Open a terminal window in Kali Linux and navigate to the `/home/kali/Desktop/imagefiles` directory.
 - Run the following:
 - `md5sum "PetersIphoneimage.zip" > peter.original.md5log.txt`
 - `sha256sum "PeterIphoneimage.zip" > peter.original.sha256log.txt`
- The file has a zip extension.
 - Note that there is another iPhone image for Rosie's phone that you will be analyzing later in the week.
 - Run the following:
 - `unzip PetersIphoneImage.zip -d PeteriPhoneImage`
 - The above command will extract the contents of the zip file and place them into a new directory called `PeteriPhoneImage`.

The Autopsy Workflow

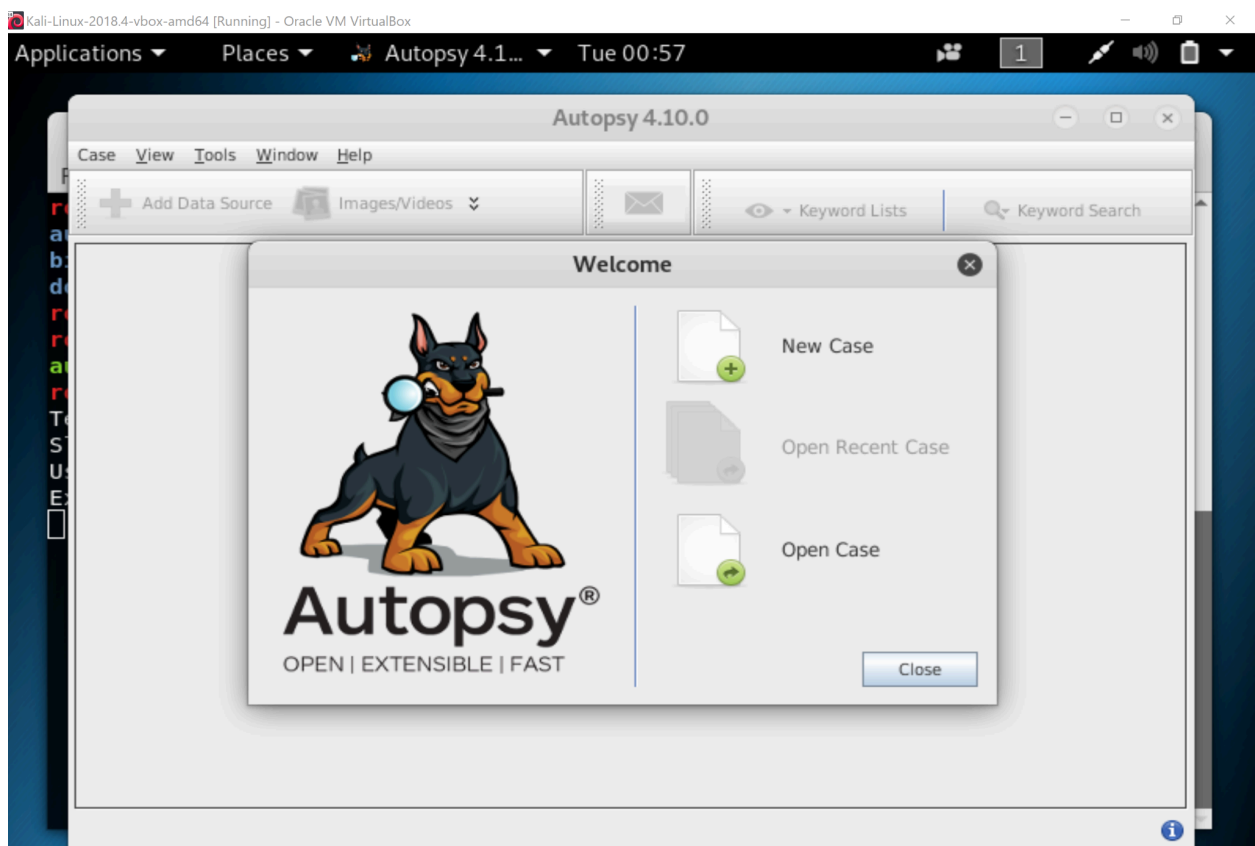
Autopsy software has a lot of features, but this demonstration will cover the following steps in the workflow:

1. Create a case: Create a case name, investigator information, and optional information. Case information helps investigators track the progress of the case.
2. Add an image: Autopsy supports Raw, EnCase, and Virtual Disk image formats. For this case, our evidence was imaged with EnCase software.
3. Configure ingest modules: For example, Email Parser, Embedded File Extractor, Android Analyzer. Ingest modules help label and categorize evidence during the ingestion process.
4. Ingest in progress: This is the process of loading the file into Autopsy. Autopsy analyzes the file and directory structure of the iPhone image, and then attempts to recreate the directory tree in its original format.
5. Manual analysis: Analyze data. During this stage, we manually research the iPhone's entire file system in search of all relevant information to the case.
6. Create timeline: Determine times, data, and data sources. During this phase, we attempt to recreate a timeline of events that the investigation team can use to understand the actions of everyone involved.

7. Report: Finally, we will consolidate all documented evidence records into a single document that will serve as evidence in a court of law. The format is either HTML or Excel.

Step 1. Create a Case

1. Navigate to your desktop and double-click on the Autopsy Icon.
2. In the Welcome window, select New Case.



3. Type the Case Name: Pure-Gold-Credit-Union.
4. Type the Base Directory: /home/kali/caseFiles/.
5. For Case Type, select Single-user.
6. Click Next.

New Case Information

Steps

1. Case Information
2. Optional Information

Case Information

Case Name:

Base Directory:

Case Type: ☒ Single-User ☐ Multi-User

Case data will be stored in the following directory:

< Back Next > Finish Cancel Help

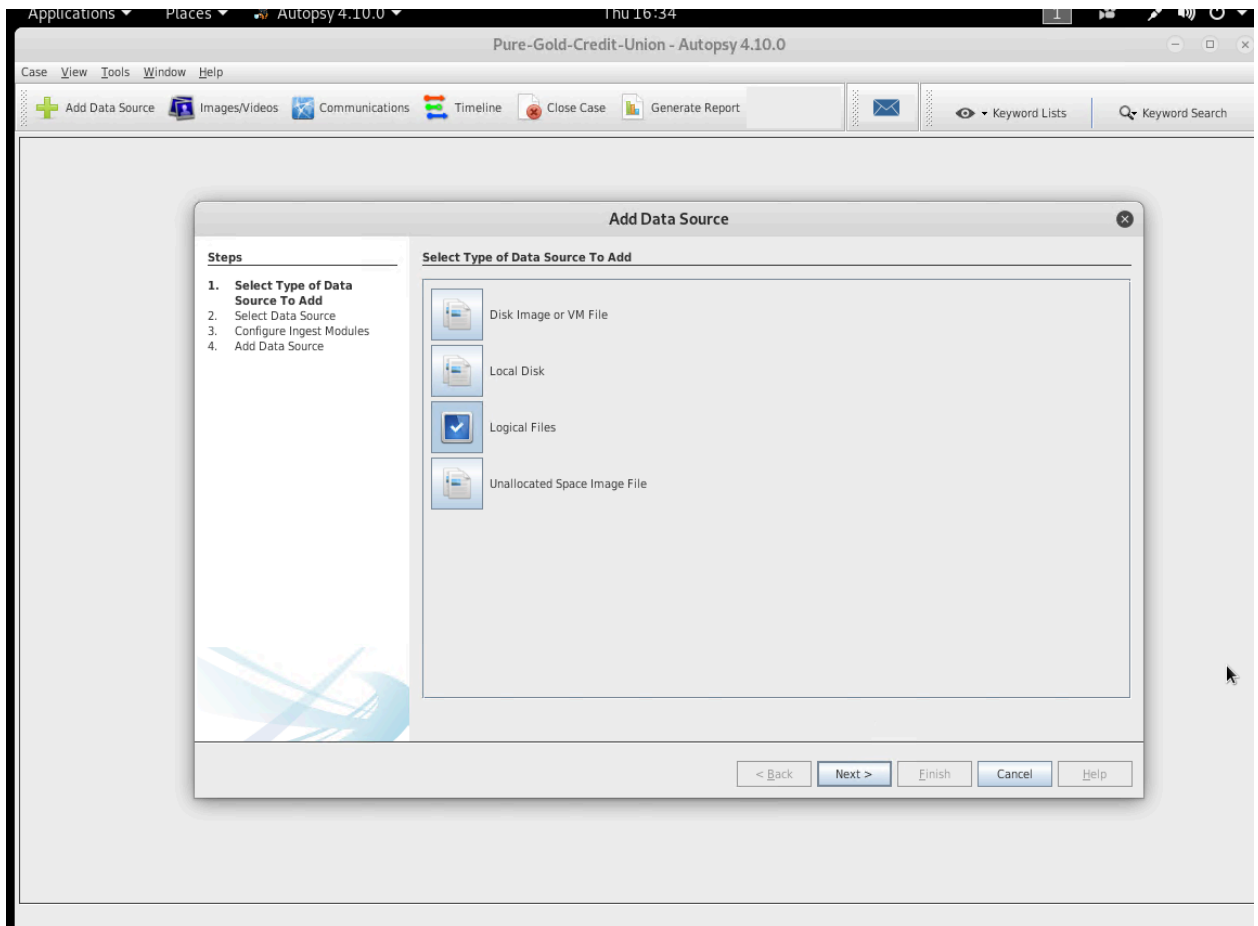
Next, enter the optional information:

- Enter a Case Number relevant to your case. For this demo, we'll use 1EZ215-P.
 - Case names and numbers help investigators keep track of the hundreds or thousands of cases that may be in the system.
- We'll leave the other optional information blank: Name, phone, email, and notes.
- Select Finish.

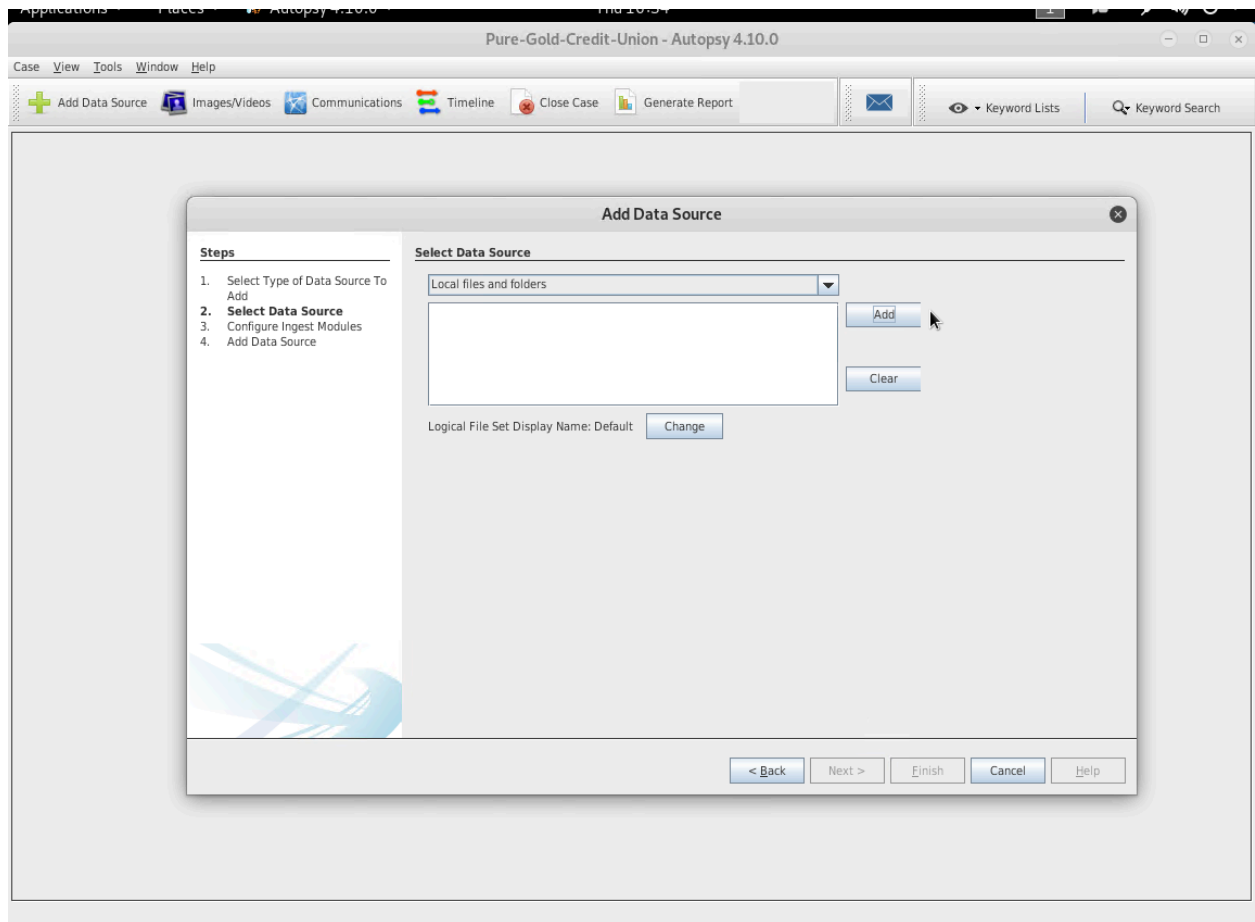
Next, we'll add the evidence file and configure ingest modules.

Step 2: Add an Image

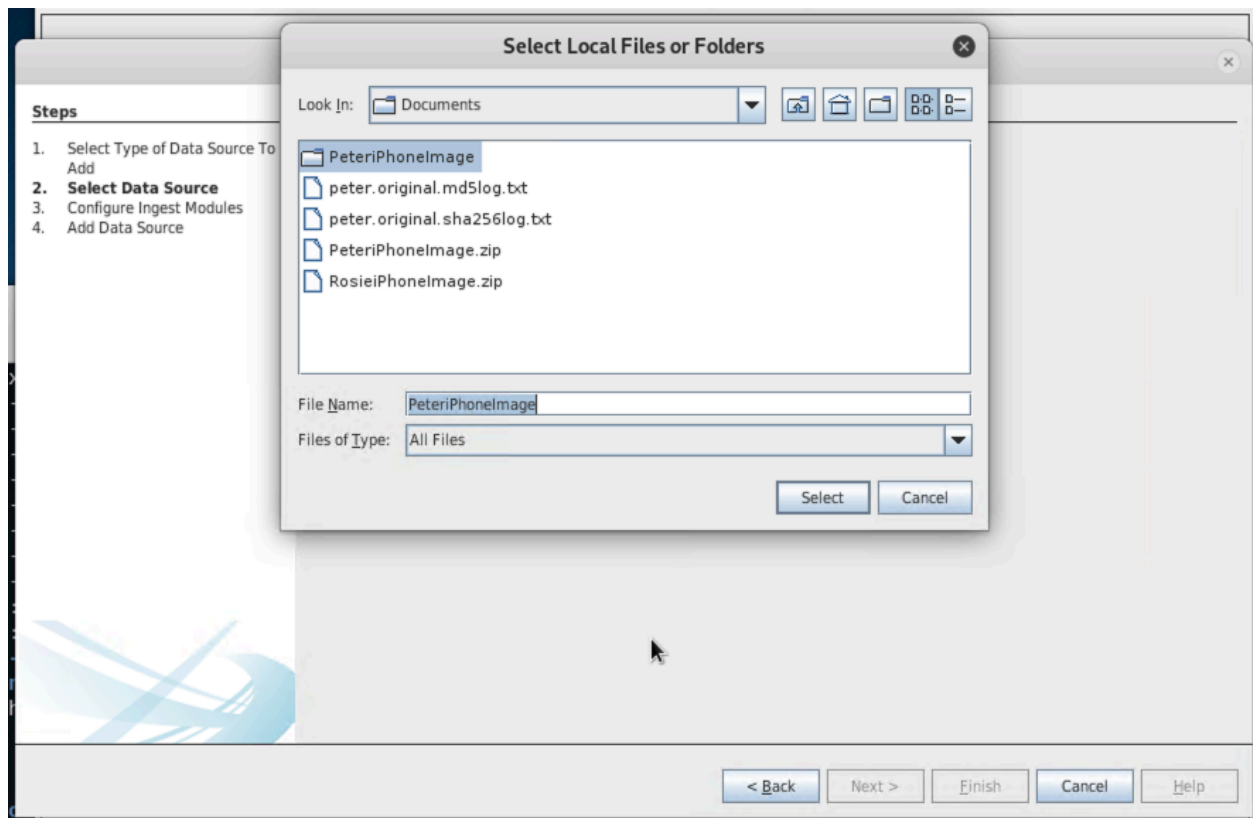
1. On the first page, leave the default option Generate new host name based on data source name, then select Next
2. Select Logical Files.
3. Click Next.



4. Leave the dropdown as "Local Files and Folders", then click "Add".



5. In the pop-up window, navigate to the location where you unzipped Peter's iPhone, and select the directory named PeteriPhoneImage, and click Select.



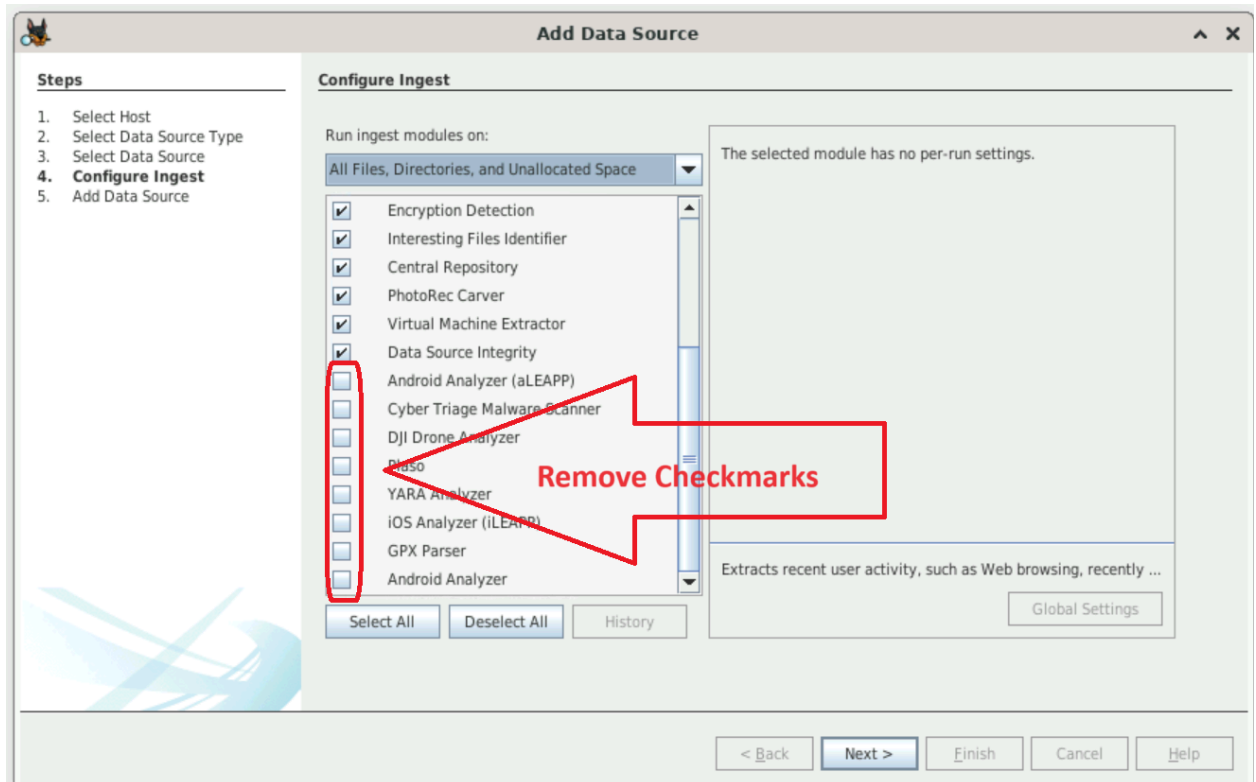
6. Click Next and then proceed to Step 3.

Step 3: Configure Ingest Modules

Ingest modules help label and categorize evidence during the ingestion process.

We'll need to remove several ingest modules on this step, as they are not relevant to our case.

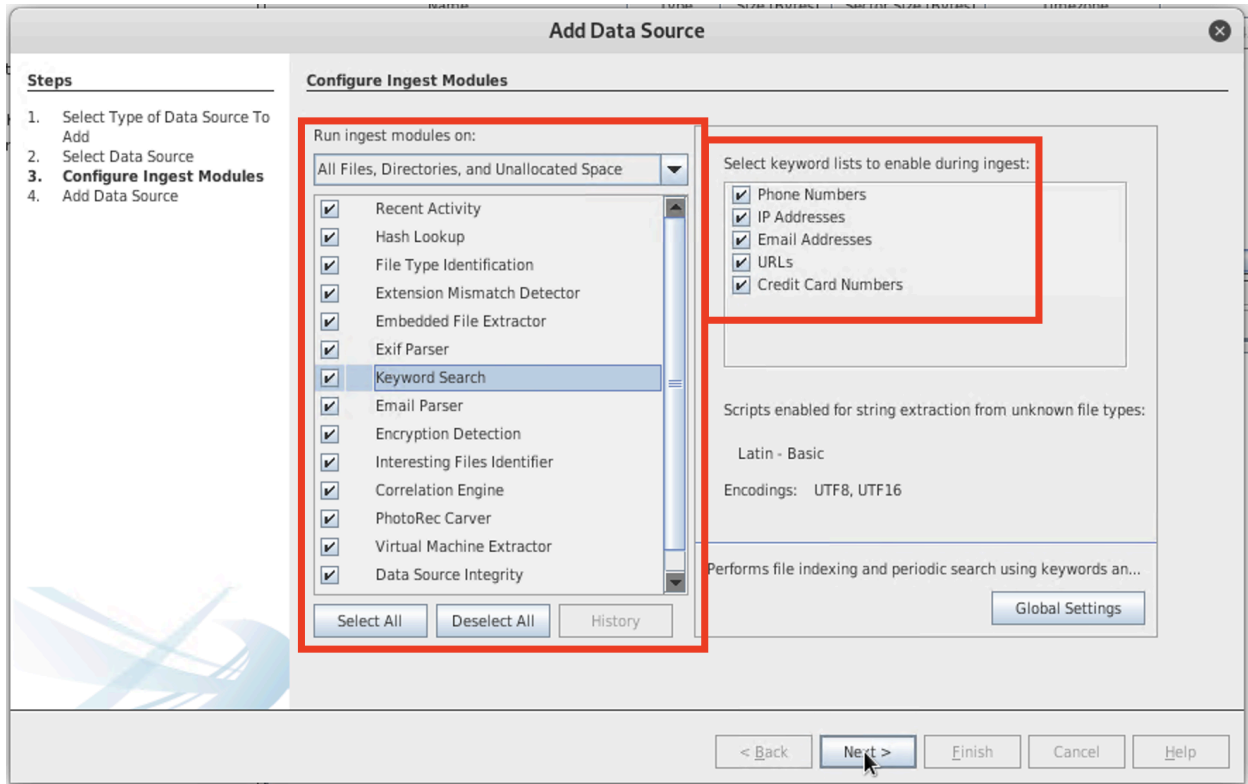
- Remove all the checkboxes below Data Source Integrity



- Next, Click Keyword Search and check all of the boxes.

Two ingest modules are particularly important: Extension Mismatch Detector and Keyword Search.

- Extension Mismatch Detector uses the results from the File Type Identification and flags files that have an extension not traditionally associated with the file's detected type. In other words, it ignores "known" files.
- Keyword Search is used to extract the maximum amount of data from the files being indexed.
 - The indexing will try to extract data from supported file formats, such as text files:
 - Phone numbers
 - IP addresses
 - Email addresses
 - URLs
 - Credit card numbers

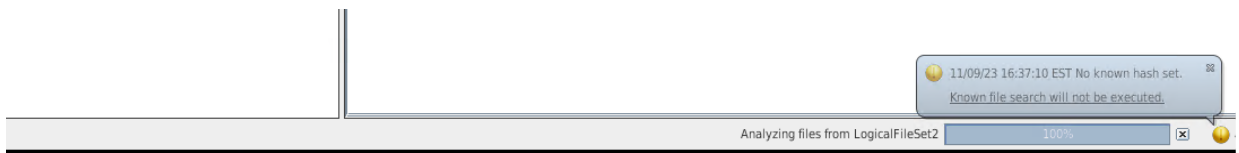


- Ensure that the Extension Manager and Keyword Search boxes are checked and click Next.

Step 4: Ingest in Progress

- Autopsy is now analyzing the image file. When this process is finished, the file is ready to be viewed and analyzed by the investigator.

The status is shown in the bottom-right corner of the window. You can click the progress bar to see the status.



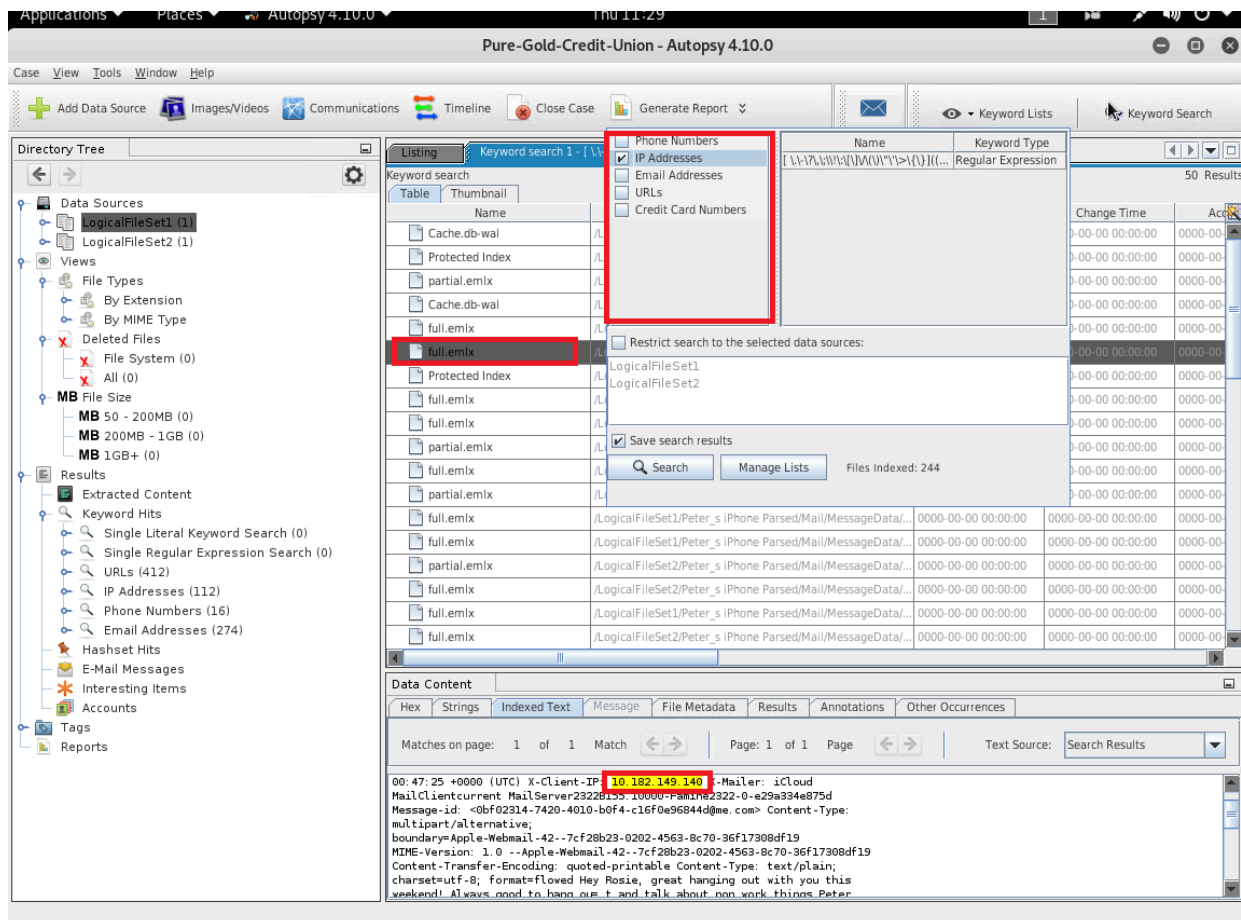
10. Break

11. Autopsy Continued

Step 5: Manual Analysis with Keyword Search

We can see the results of the Keyword Search ingest module as follows:

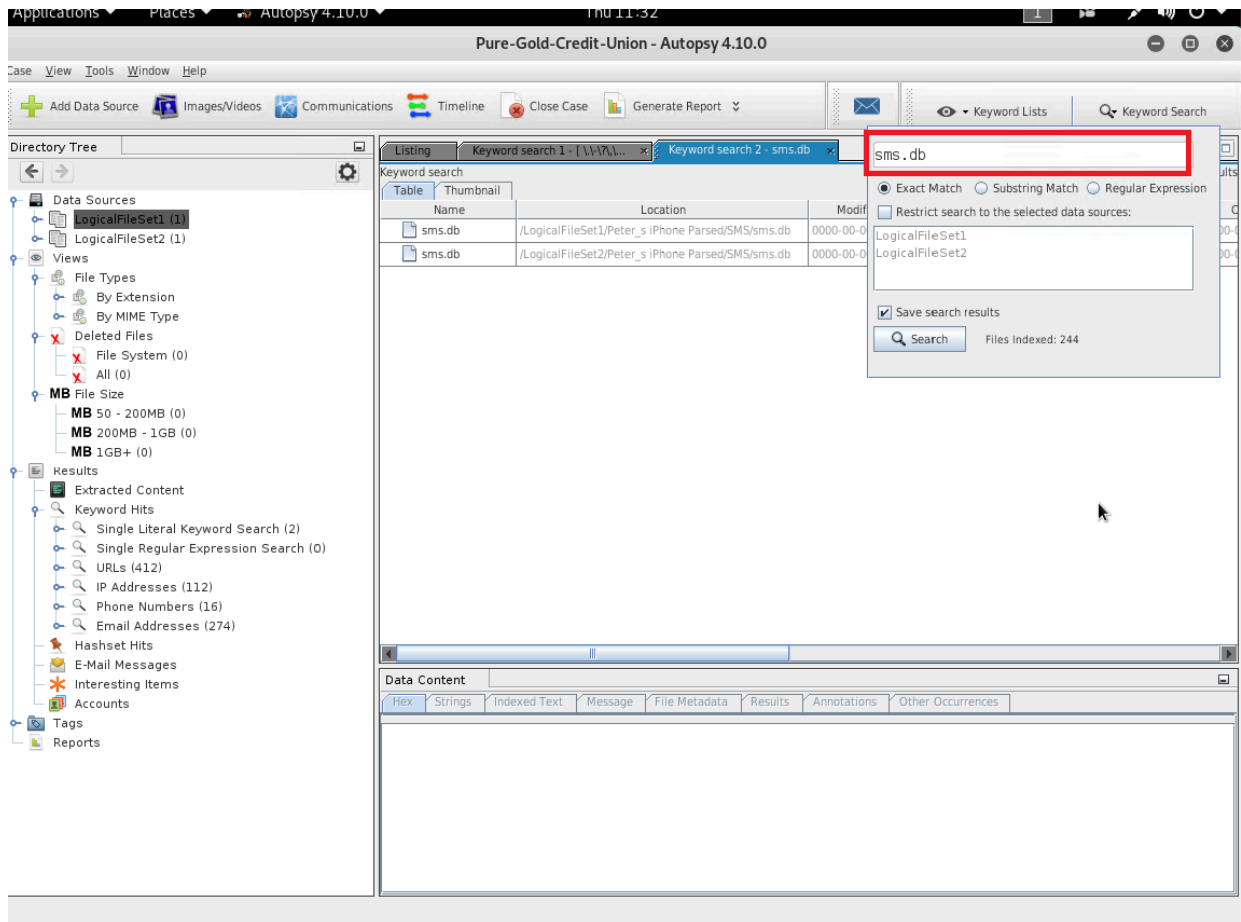
- Click on Keyword Lists in the upper-right portion of the toolbar.
 - We can see all of the categories that Autopsy discovered during the file ingest process, such as phone numbers and IP addresses, etc.
- In the dropdown, check IP Addresses.
- Click Search.
- Autopsy will highlight all of the IPs contained in the file for easy reference.



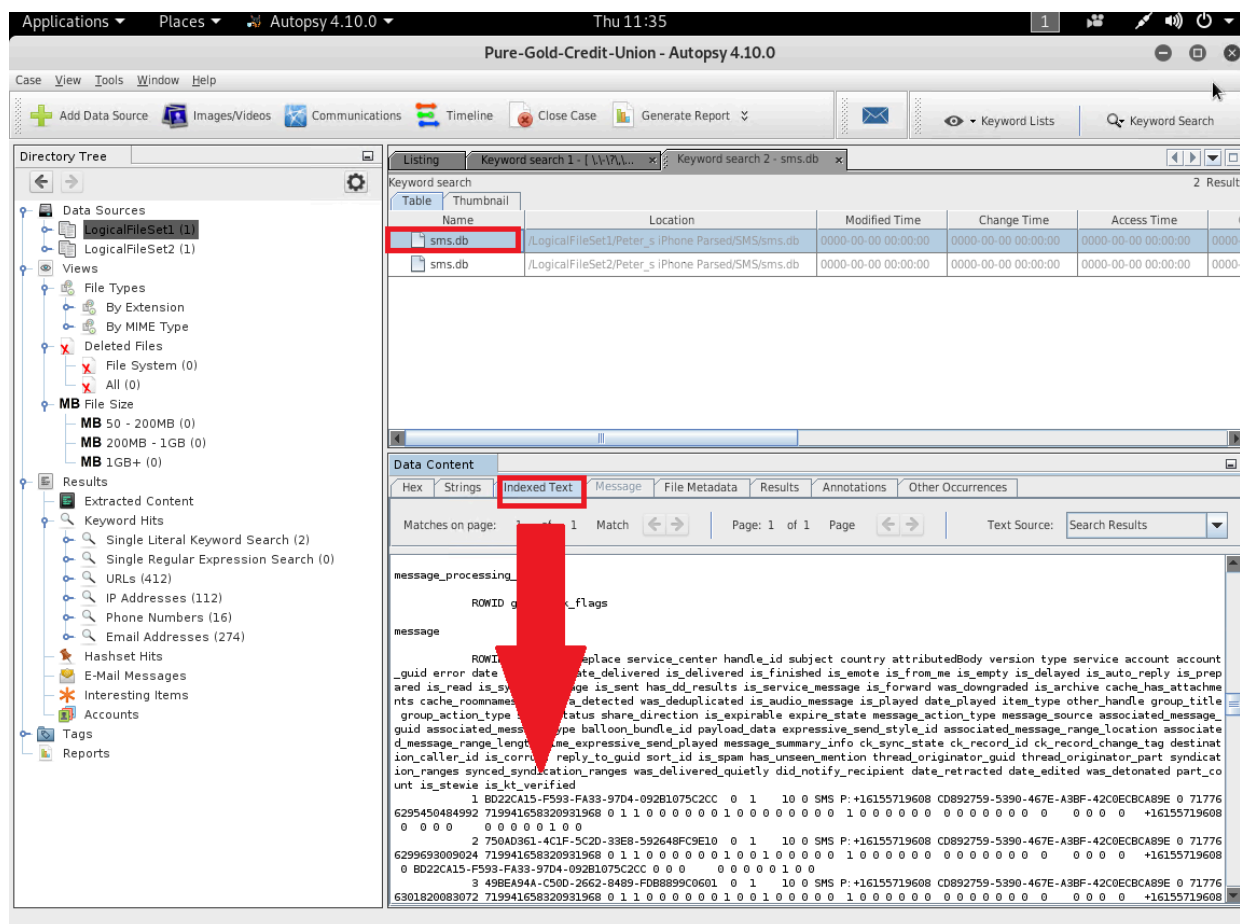
Step 5b: Manual Analysis of SMS Messages

Autopsy provides us with a way to view and read SMS messages:

- Click the Keywords Search box in the upper-right corner.
 - In the dropdown, enter `sms.db` and click Search.



- You should now see the `sms.db` file populated in the Keyword Search (the lower-right window pane) in the Table tab.
- Click the `sms.db` file to highlight it.
 - You should now be able to read SMS messages in the Data Content window pane in the lower-right window inside the Indexed Text tab.



- What types of information can SMS messages reveal?
 - Answer: SMS messages can implicate other individuals involved in the crime. They can also reveal the objective of the crime or the motives for committing it.
- In the next class, we will examine an iPhone image. We'll continue to learn about reporting in future classes.

12. Wrap-Up and Review

Digital forensics relies on the expertise of examiners to analyze and interpret data retrieved by trusted forensics examination tools.

- You can perform a complete investigation using only open-source tools.
- Autopsy is an open-source, multi-platform, graphical tool that runs on Windows, Ubuntu, Kali Linux, and OS X platforms.

The steps we needed to complete for our first autopsy investigation were:

- Generate MD5 and SHA-256 hashes for the evidence to ensure the integrity of the evidence.
- Specify a case name and number to help investigators keep track of the files and their progress.
- Select the file type to be ingested, i.e., .zip, and specify a working directory where progress will be saved.
- Specify a standard time zone for the case.
- Configure ingest modules to help categorize, label, and organize data.

Congrats on a job well done starting the investigation!

In the next class, we'll continue our investigation using Autopsy to analyze artifacts on iPhone images.